Reconstructing a Weak-Lensing Radial-Acceleration Relation from Public Object-Level Data:

Spin-2 Orientation, Additive-Control Contamination, and a Paired-Covariance Gate

Facundo Firmenich[1]

[1]Centro de Estudios del Sur (CEDESUR), Universidad Nacional Arturo Jauretche, Buenos Aires, Argentina

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Abstract

Independent reconstruction of weak-lensing radial-acceleration relations (RARs) requires more than reproducing a published curve: the lens selection, spin-2 orientation, source–lens pairing, random-centre subtraction, deprojection, and covariance must share one auditable estimator. We report an object-level reconstruction built from public KiDS-1000 products. The pipeline scans 21,262,011 source rows, exactly recovers the declared 106,843-lens selection, and accumulates 3,017,858,899 accepted source–lens pairs. A preregistered four-convention diagnostic localizes the dominant historical discrepancy to the catalogue e_2 sign convention in the right-ascension/declination frame. The corrected estimator reduces the median absolute log discrepancy from 1.03654 to 0.04943 dex, retains 15/15 positive RAR bins, and keeps the largest cross-channel residual below 1.76 diagonal standard deviations. It nevertheless fails the frozen reproduction gate because only 10/15 deproject-first central values lie within the published total 1σ envelopes. A real stratified pilot using 10,060 random centres and 433,841,322 accepted pairs finds a material large-scale tangential control; in the outermost bin its subtraction changes the inferred sign. The pilot cannot replace the frozen 45,038,900-centre control. We therefore formalize the next estimand as $D = S - R$ with matched delete-one spatial replicates and covariance $C_D = C_S + C_R - C_{SR} - C_{SR}^{\mathsf{T}}$. The present result is an estimator audit and replication boundary, not a dark-matter, modified-gravity, or plasma detection.

Keywords: weak gravitational lensing; galaxy–galaxy lensing; radial acceleration relation; reproducibility; covariance; survey systematics

1 Introduction

Galaxy–galaxy weak lensing measures the coherent distortion of background source shapes around a foreground lens population. At the required precision, the observable is inseparable from source calibration, angular conventions, survey geometry, redshift weighting, additive controls, and covariance construction [Mandelbaum, 2018, Singh et al., 2017]. Radial-acceleration relations extend this inference by converting an excess-surface-density profile into an effective radial acceleration and comparing it with the acceleration predicted from baryonic mass.

KiDS-1000 has supplied both public weak-lensing products and published RAR analyses [Kuijken et al., 2019, Wright et al., 2020, Brouwer et al., 2021]. A later joint kinematic–lensing analysis introduced an exact deprojection formula and reported a smooth continuation of the kinematic RAR over roughly

2.5 dex in acceleration [Mistele et al., 2024]. These results motivate a demanding reproducibility question: can the RAR be reconstructed from public object-level inputs without importing the target curve into the estimator?

The contribution here is not a new physical ontology for the mass discrepancy. It is a staged reconstruction in which adverse results are retained. The pipeline first establishes byte-level custody and exact sample selection, then executes the full source–lens accumulation, diagnoses orientation with rules fixed before comparing the four candidates, measures a real random-centre pilot, and finally freezes the paired-covariance gate required for a corrected RAR. This separation prevents successful software tests or a visually similar curve from being relabelled as physical confirmation.

2 Data and jurisdiction

2.1 Public source and lens products

The source surface is the KiDS-1000 SOM-gold weak-lensing catalogue. Its 17,712,469,440 bytes are represented by 185 contiguous remote byte ranges so that the full file need not be materialized on a workstation. The complete scan covers 21,262,011 rows. Published quality cuts retain 19,109,925 rows with finite positive shape weights. The five photometric-bin counts are 1,727,608; 3,442,980; 5,624,220; 4,006,695; and 4,308,422.

Native GAAP u- and r-band photometry was aligned object by object for 1,239,422 KiDS-bright records. Applying the declared isolation and selection geometry exactly recovers 106,843 lenses, with zero count difference and no reconstructed-magnitude fallback. The official five SOM $n(z)$ distributions were integrated behind the lens posterior to form an effective critical surface-density lookup spanning 3.047×10^{15}–$9.010 \times 10^{15}\, M_\odot\,\mathrm{Mpc}^{-2}$.

2.2 Stage authority

Table 1 records what each completed stage can establish. A stage may validate an input or estimator while remaining explicitly non-authoritative for a physical claim.

Table 1: Authority of the executed reconstruction stages.

Stage	Completed evidence	Claim not licensed
v0.16	Exact lens selection; full source attrition; published-bin RAR transform	Object-level lensing reproduction or mechanism
v0.17 signal	21.26 million sources, 106,843 lenses, 3.018 billion accepted pairs	Corrected survey estimator without random subtraction
v0.17 orientation	Frozen comparison of four spin-2 conventions	Final reproduction or model selection
v0.17 random pilot	10,060 real random centres; 433.84 million accepted pairs	Frozen 45.04 million-centre control
v0.18 gate	Input integrity and paired-covariance implementation tests	New corrected RAR or scientific success

3 Estimator

3.1 Spin-2 rotation

For a source ellipticity (e_1, e_2) and a source–lens polar angle ϕ measured from east, the repaired KiDS right-ascension/declination convention is

$$e_t = -\left[e_1 \cos(2\phi) - e_2 \sin(2\phi)\right], \tag{1}$$

$$e_\times = e_1 \sin(2\phi) + e_2 \cos(2\phi). \tag{2}$$

The four-candidate diagnostic crossed two angular origins with the documented e_2 sign transformation. Candidate definitions, the pass rule, and the published comparison were frozen before adjudication. The implementation retains the four linear basis accumulators, so conventions can be evaluated without rereading the 17.7 GB source surface or adaptively rotating the data.

3.2 From excess surface density to acceleration

Each accepted lens–source pair contributes weighted sufficient statistics in 27 projected-radius bins. Source redshift uncertainty enters through the effective critical surface density. The deproject-first channel applies the signed linear deprojection operator to each lens profile before stacking, following the exact transformation in Mistele et al. [2024]; it is kept distinct from a stack-first singular-isothermal-sphere diagnostic.

The full signal scan was split into eight contiguous source-row partitions. Every partition preserves the lens payload, radial edges, and critical-density lookup hashes. The merged artifact covers $[0, 21{,}262{,}011)$ without a gap and has content SHA-256 prefix `5d5f8c3a11cbe99b`; the complete digest is retained in the evidence ledger distributed with the package.

3.3 Corrected estimand and covariance

Let S denote the signal-centre estimator and R the random-centre estimator. The survey-corrected quantity is

$$D = S - R. \tag{3}$$

If both populations reuse source galaxies, independence cannot be assumed. Matched spatial delete-one replicates give

$$C_D = C_S + C_R - C_{SR} - C_{SR}^{\mathsf{T}}. \tag{4}$$

The implementation operates on region-level additive sufficient sums, deletes the same region from S and R, and builds one joint vector containing tangential, cross, deproject-first tangential, and deproject-first cross channels. A coordinate is estimable only when finite in the full estimator and every matched replicate. Pairwise deletion, zero filling, diagonal substitution, automatic inversion, and a silent Hartlap correction [Hartlap et al., 2007] are prohibited.

4 Results

4.1 Full object-level signal scan

The complete eight-partition scan evaluates all 21,262,011 source rows against the exact 106,843-lens payload and accepts 3,017,858,899 pairs. The merged sufficient-statistics surface is compact (138.6 MB across the eight sealed artifacts) because individual pairs are never persisted. Continuous row coverage and content hashes separate the observational result from transport and reporting failures.

4.2 Orientation repair

Figure 1 shows the only preregistered repair candidate. The catalogue-e_2 sign transformation reduces the deproject-first median absolute log discrepancy from 1.0365400070 to 0.0494255413 dex, an improvement of 0.9871144657 dex. All 15 central values are positive, and the largest cross-channel residual is 1.75594 diagonal standard deviations. The independent stack-first diagnostic gives a 0.0436427705 dex median discrepancy and 15/15 positive bins.

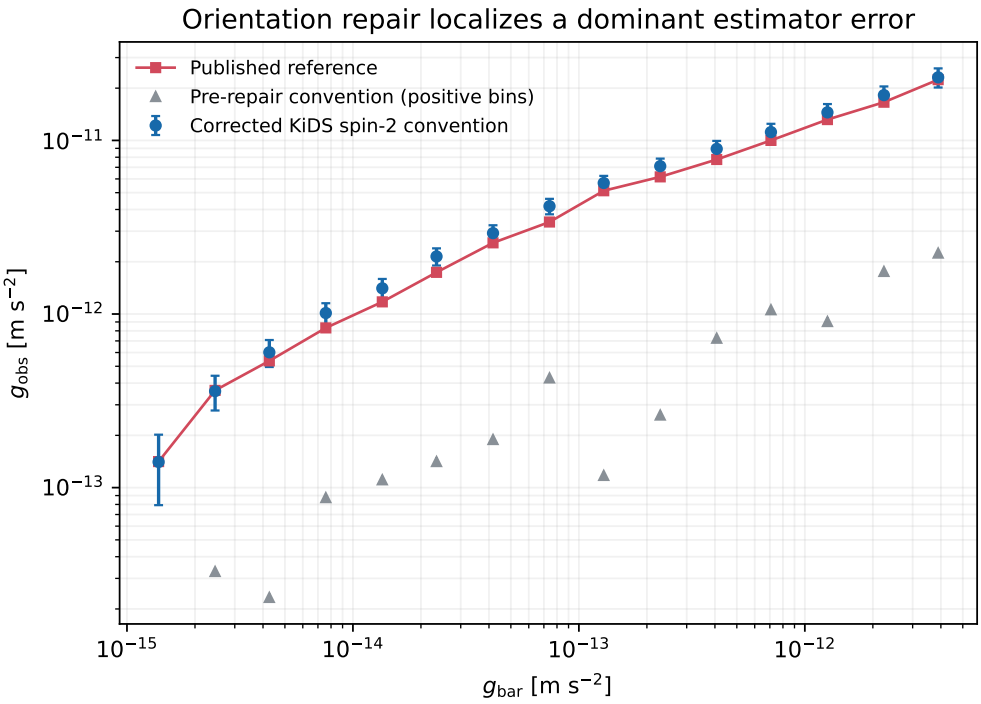


Figure 1: Object-level RAR after the frozen spin-2 orientation adjudication. Error bars are the available diagonal uncertainties; they are not a substitute for the paired spatial covariance in Eq. (4).

The repair is substantial but not a completed reproduction. Only 10/15 deproject-first central values and 12/15 stack-first values fall within the published total 1σ envelopes. The frozen rule therefore returns `reproduction_gate=false`. That adverse decision is preserved rather than weakening the threshold after seeing the improved curve.

4.3 A real random-centre pilot

The random pilot uses ten positions in each of the 1,006 official survey tiles, for 10,060 centres total. Against the same 21,262,011-source surface, it evaluates 575,059,669 candidate pairs and accepts 433,841,322. The largest cross-channel diagnostic stays below 1.87 diagonal standard deviations.

The tangential random surface is nevertheless material. At $R = 10.24094\,h_{70}^{-1}$ Mpc,

$$\Delta\Sigma_R = (-2.8076 \pm 0.9209) \times 10^{11}\,M_\odot\,\mathrm{Mpc}^{-2}, \tag{5}$$

$$g_{\mathrm{obs},R} = (-1.5591 \pm 0.3120) \times 10^{-13}\,\mathrm{m\,s}^{-2}, \tag{6}$$

using diagonal errors. These are localization diagnostics, not global significances. At the same radius, subtracting the pilot from the signal moves $g_{\rm obs}$ from -5.40×10^{-14} to $+1.02 \times 10^{-13}\,\mathrm{m\,s}^{-2}$.

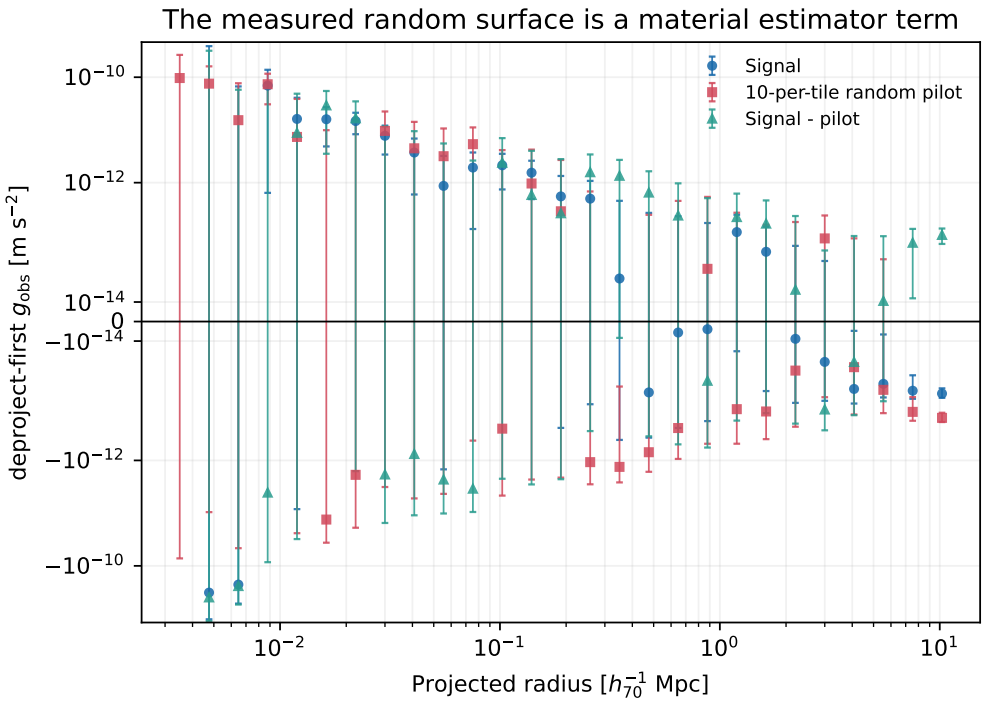


Figure 2: Full signal, measured stratified random pilot, and their exploratory difference. The signed logarithmic ordinate preserves negative values. The pilot demonstrates estimator materiality but cannot replace the frozen 50x control.

4.4 Why the result stops here

The frozen random catalogue contains 45,038,900 positions, 4,477.03 times the pilot count. Linear extrapolation from the original implementation projects roughly 2.57×10^{12} candidate and 1.94×10^{12} accepted pairs, with 231–417 CPU hours per original partition. This estimate is a planning bound, not a benchmark of the tile-local successor.

The v0.18 input audit passes all four frozen-file hashes, exact coverage of 21,262,011 source rows, 988 source pointings, 1,006 random tiles, and 45,038,900 allocated random positions. Seventeen focal algebra/provenance tests pass. The terminal status remains `BLOCKED_EXECUTION_AND_PAIRED_COVARIANCE` because the full control, matched signal/random spatial replicates, and an admitted finite compute budget do not yet exist.

5 Interpretation and limits

Three conclusions are supported. First, a sign convention in a spin-2 observable can dominate a full-catalogue reproduction while leaving transport, selection, and deprojection machinery internally consistent. Second, the measured random-centre surface is not a cosmetic correction: at large radius it is comparable to or larger than the signal and can reverse its sign. Third, a random-only jackknife estimates C_R, not the covariance of $S - R$; shared sources require the cross terms in Eq. (4).

The current evidence does not adjudicate ΛCDM, modified gravity, or a plasma/morphotopological mechanism. It does not establish a final corrected RAR, a global p-value, or a total-systematics covariance. Remaining uncertainties include shear calibration, source photometric redshifts, lens mass modelling, survey selection, deprojection tails, field variance, and the design of comparable-area spatial

exclusions. The 1,006 raw survey tiles are not automatically valid equal-area jackknife regions.

6 Confirmatory next experiment

The decisive run has a fixed sequence:

1. bind a remote node with at least 40 GB scratch and a finite CPU-hour budget;
2. execute a hard 30-minute tile-local benchmark and seal its time/memory/pair/byte receipt;
3. freeze a justified spatial exclusion map and source-deletion policy;
4. accumulate both signal and all 45,038,900 random centres into the same region-level sufficient statistics;
5. form matched replicates, $D = S - R$, and the full four-channel covariance;
6. apply the unchanged reproduction gate before any mechanism-level interpretation.

No step authorizes the next automatically; every result, including failure or singular covariance, is a valid terminal scientific outcome.

7 Conclusions

The public-data reconstruction has crossed the expensive object-level threshold: exact lens selection, the complete source scan, more than three billion accepted signal pairs, a preregistered orientation repair, and a real random-centre pilot. The repaired signal is close to the published RAR in median log discrepancy but fails the stricter all-bin reproduction rule.

The most important empirical result is the boundary itself. A measured large-scale random term is capable of materially changing the derived acceleration surface, while the covariance of the corrected estimator requires matched signal–random spatial replicates. Until the frozen 50x control and Eq. (4) are executed, a final RAR or physical-model verdict would outrun the data.

Data and code availability

The executed code, frozen protocol, compact sufficient-statistics receipts, and evidence files are versioned in the public repository `https://github.com/FacundoFirmenich/darkpipe-realdata`. The governing working branch is `v018-paired-random-covariance-gate` at commit `df321b1f`; draft pull request 25 records the gate and adverse CI history. The raw KiDS products remain at their official providers and are not duplicated in this paper package.

Acknowledgements

This work uses public KiDS-1000 products. Any submitted version must include the complete KiDS acknowledgement and programme identifiers required by the data provider, together with the final funding, conflict, CRediT, ORCID, and AI-assistance declarations approved by the author.

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A Compact evidence ledger

Artifact	Git blob SHA	Governing content
V016 critical checkpoint	`a07fe9d6...`	Exact 106,843-lens closure; 21,262,011-source scan; effective critical-density lookup.
Orientation metrics	`b8097436...`	Four frozen conventions; repaired median discrepancy; all-bin gate.
Partition custody	`43cd6f39...`	Eight complete partitions and 3,017,858,899 accepted pairs.
V017 random comparison	`7d091464...`	10,060-centre pilot; material tangential random surface.
`readiness.json`	`224855ef...`	Passing input integrity and explicit three-blocker terminal gate.